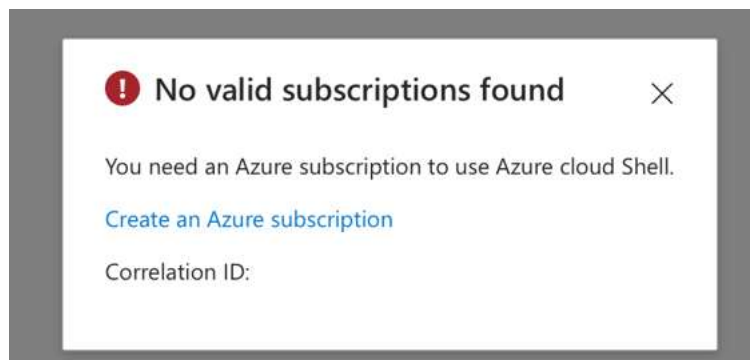


## Task 'Unit 4: Cloud Native Technology Part 1'

### Azure resources using a Bash provisioning

1. I deployed and configured resources successfully on Azure using a Bash provisioning script that was executed in Azure CLI. An active Azure subscription must be enabled before as a pre-requisite.

**Figure 1:** Subscription not Found and must be enabled Before



2. I signed into the Azure cross-platform command-line interface using `az login` and checked the active subscription with `az account show`.
3. I created a new file of the using `nano azure-provision.sh`. and pasted the content of provisioning script, then made the script executable and `chmod +x azure-provision.sh` and run the bash using `./azure-provision.sh`. The cloud-demo-rg resource group and the cloud-demo-vm virtual machine have been deployed. I ran into some minor issue that I was able to quickly resolve as shown in the screenshots.

**Figure 2:** Azure Script Deployment Preparations

## Task 'Unit 4: Cloud Native Technology Part 1'

```

C:\Users\Thomas> Runas /user:Administrator powershell
PowerShell v5.1.10.0 [Administrator]
PS C:\Users\Thomas> .\az login
Connecting to Azure Cloud Shell...
Welcome to Azure Cloud Shell
Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell
Subscription used to launch your CloudShell [redacted] is not registered to Microsoft CloudShell NameSpace. Please follow these instructions "https://aka.ms/Regi
sterCloudShell" to register. In future, unregistered subscriptions will have restricted access to CloudShell service.
Your Cloud Shell session will be ephemeral as no files or system changes will persist beyond your current session.
azure | ~ | az az login
azure | ~ | az az login
azure | ~ | az az login
Cloud Shell is automatically authenticated under the initial account signed-in with. Run "az login" only if you need to use a different account
to sign in. Use a web browser to open the page https://login.microsoft.com/device and enter the code QJ88EM5E to authenticate.
Retrieving tenants and subscriptions for the selection...
The following tenants don't contain accessible subscriptions. Use "az login --allow-ss-subscriptions" to have tenant level access.
No subscriptions found for tenant [redacted]
azure | ~ | az az login
Cloud Shell is automatically authenticated under the initial account signed-in with. Run "az login" only if you need to use a different account
to sign in. Use a web browser to open the page https://login.microsoft.com/device and enter the code QJ88EM5E to authenticate.
Retrieving tenants and subscriptions for the selection...
[tenant and subscription selection]
No Subscription name Subscription ID Tenant
[1] * Azure subscription 1 [redacted]
The default is marked with an *: the default tenant is [redacted] and subscription is 'Azure subscription 1' [redacted].
Select a subscription and tenant (Type a number or enter for no changes): 1
Tenant: [redacted]
Subscription: Azure subscription 1 [redacted]
[AccessComplete]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?link=1
If you encounter any problem, please open an issue at https://aka.ms/azcliBog
[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.
azure | ~ | az az account show
{
  "environmentName": "AzureCloud",
  "homeTenantId": "[redacted]",
  "id": "[redacted]",
  "isDefault": true,
  "managedByTenants": [],
  "name": "Azure subscription 1",
  "state": "Enabled",
  "tenantDefaultDomain": "[redacted]",
  "tenantDisplayName": "[redacted]",
  "tenantId": "[redacted]",
  "user": {
    "name": "[redacted]@microsoft.com",

```

### Figure 3: Azure Script Deployment Execution

```

amon@ ~ % ./azure-provision.sh

{"id": "/subscriptions/xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx/resourceGroups/cloud-demo-rg",
 "location": "eastus",
 "managedBy": null,
 "name": "cloud-demo-rg",
 "properties": {
   "provisioningState": "Succeeded"
 },
 "tags": null,
 "type": "Microsoft.Resources/resourceGroups"}

SSH key files "/home/amon/.ssh/id_rsa" and "/home/amon/.ssh/id_rsa.pub" have been generated under ~/.ssh to allow SSH access to the VM. If using machines without permanent storage, back up your keys to a safe location.
Consider upgrading security for your workloads using Azure Trusted Launch VMs. To know more about Trusted Launch, please visit https://aka.ms/TrustedLaunch.

amon@ ~ % as provider register --namespace Microsoft.Compute
Registering is still on-going. You can monitor using 'az provider show -n Microsoft.Compute'

amon@ ~ % as provider show \
--namespace Microsoft.Compute \
--query registrationState \
--output text
Registered
amon@ ~ % ./azure-provision.sh

{"id": "/subscriptions/xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx/resourceGroups/cloud-demo-rg",
 "location": "eastus",
 "managedBy": null,
 "name": "cloud-demo-rg",
 "properties": {
   "provisioningState": "Succeeded"
 },
 "tags": null,
 "type": "Microsoft.Resources/resourceGroups"}

Consider upgrading security for your workloads using Azure Trusted Launch VMs. To know more about Trusted Launch, please visit https://aka.ms/TrustedLaunch.

```

4. I verified the deployed resources using:

```
az resources list -g cloud-demo-rg -o table
```

## Task 'Unit 4: Cloud Native Technology Part 1'

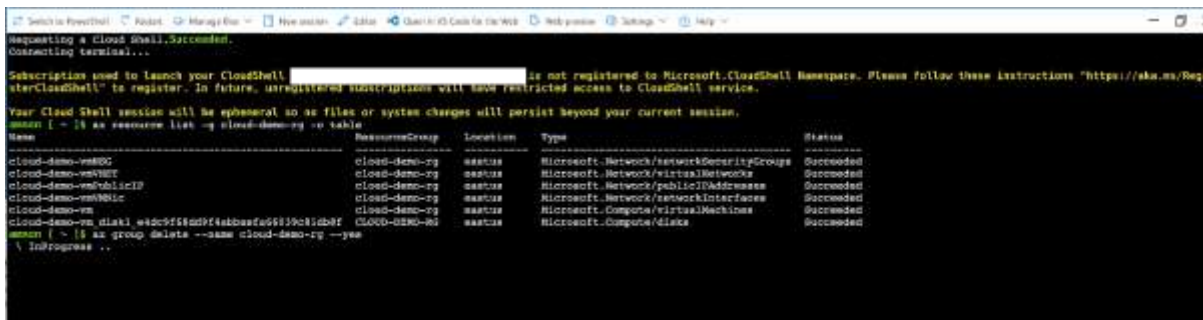
5. After testing the deployment, I verified the list resource group using:

```
az group list --name cloud-demo-rg --yes
```

Finally completed the operation by deleting the resources group using the: `az group delete --name cloud-demo-rg --yes`

This was done to remove the resources and avoid unnecessary Azure costs.

**Figure 4: Azure CLI Resources Listing and Deletion**



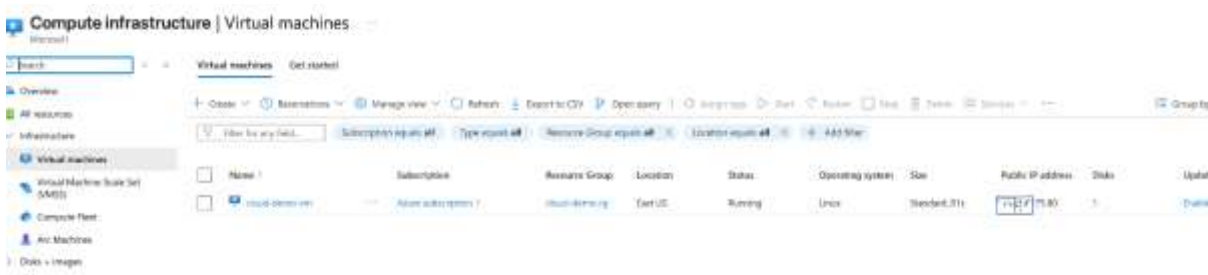
```
Switch to PowerShell | Azure | Manage files | View console | Edit | Open in VS Code for the Web | Web preview | Settings | Help
Requesting a Cloud Shell. Succeeded.
Connecting terminal...

Subscription used to launch your CloudShell is not registered to Microsoft.CloudShell Namespace. Please follow these instructions "https://aka.ms/RequesterCloudShell" to register. In future, unregistered subscriptions will have restricted access to CloudShell service.

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
az group list --name cloud-demo-rg --yes
Name                                ResourceGroup    Location    Type                                Status
-----
cloud-demo-vnet                     cloud-demo-rg    eastus     Microsoft.Network/virtualNetworks  Succeeded
cloud-demo-vnetSubnet                cloud-demo-rg    eastus     Microsoft.Network/subnets          Succeeded
cloud-demo-vnetPublicIP              cloud-demo-rg    eastus     Microsoft.Network/publicIPAddresses Succeeded
cloud-demo-vnetNATRule               cloud-demo-rg    eastus     Microsoft.Network/natRules          Succeeded
cloud-demo-vm                        cloud-demo-rg    eastus     Microsoft.Compute/virtualMachines   Succeeded
cloud-demo-vmDisk                    cloud-demo-rg    eastus     Microsoft.Compute/disks             Succeeded
az group delete --name cloud-demo-rg --yes
\ InProgress ..
```

6. I also verified The deployed resources through the Azure web portal, as shown in the screenshot.

**Figure 3: Deployed Resources on the Azure Web Portal**



## **Task 'Unit 4: Cloud Native Technology Part 1'**

### **References**

Microsoft (2026c) Az Resource Manage Azure resources.. Microsoft Learn Available at: <https://learn.microsoft.com/en-us/cli/azure/resource?view=azure-cli-latest> [Accessed: 19 August 2026].

### **Use of Digital Tools**

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